Simultaneous Orbital Decay and Precession in Exoplanet TTVs

by

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Abstract

Transit Timing Variations (TTVs) are widely used to infer orbital evolution in exoplanetary systems, typically attributing observed trends to either orbital decay or apsidal precession independently. In this work, we investigate the possibility that these effects occur simultaneously and assess their combined impact on TTV modeling and parameter inference.

We introduce a coupled decay–precession (DP) model that incorporates both orbital decay and precession within a unified framework. Using numerical simulations based on a stroboscopic method, we generate synthetic TTV signals including perturbations from orbital decay and stellar oblateness–induced precession. We explore a range of decay rates, precession strengths, orbital eccentricities, and observational baselines.

Model comparison is performed using nested fitting and Bayesian evidence to quantify the relative support for decay-only, precession-only, and coupled DP models. We find that for short observational baselines, the DP model yields only moderate evidence when decay dominates. However, as the data length increases, the DP model becomes strongly favored over single-effect models. We also identify limitations in convergence failures at high parameter values and parameter recovery accuracy arising from degeneracies between decay and precession signals.

These results demonstrate that treating orbital decay and precession as independent processes can lead to biased interpretations of TTV data, and that their simultaneous inclusion is essential for robust characterization of exoplanet orbital evolution.

Lay Summary

Astronomers study exoplanets—planets beyond our solar system—by tracking tiny variations in the timing of their orbits. These variations are usually explained by a single effect, such as orbital decay, where a planet slowly spirals toward its star, or orbital precession, where the orbit gradually shifts in orientation.

But what if both processes happen at the same time? Although this possibility has not yet been confirmed, it could make current interpretations of data misleading.

In this work, we simulate how these combined effects would appear in observations. We find that when both are present, methods that assume only one effect can give incorrect conclusions, especially with limited data. As more observations become available, the combined signal becomes clearer.

This study suggests that exoplanet systems may be more complex than commonly assumed and that more complete models are needed to understand their evolution better.

Preface

This dissertation presents original research conducted under the supervision of Dr. Aaron Boley at the University of British Columbia.

The conceptual motivation for this project arose from discussions with my supervisor regarding observed transit timing variations in specific exoplanetary systems, particularly WASP-12b and WASP-43b. These discussions led to the central research question explored in this work: whether orbital decay and apsidal precession can occur simultaneously and how such coexistence would manifest in transit timing observations.

The computational framework developed and utilized in this thesis builds upon two primary existing resources. First, the Bayesian parameter estimation and model comparison pipeline is based on the `orbdot` software package, developed by UBC PhD candidate Simone Hagey for modeling secular orbital evolution in exoplanet systems. Second, the geometric stroboscopic simulation codebase used to generate synthetic transit timing data was originally developed by Dr. Aaron Boley.

To investigate the simultaneous effects of orbital decay and apsidal precession, substantial modifications and extensions were made to both the `orbdot` framework and the underlying simulation code. These include the implementation of a combined Decay-Precession (DP) model and the integration of coupled timing signals within the fitting pipeline.

All large-scale simulation design, data generation, parameter sweeps, statistical analysis, and manuscript preparation were carried out by me under the guidance of my supervisor.

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Nomenclature

List of Symbols and Parameters

Symbol	Unit	Description
t_0	$\mathrm{BJD_{TDB}}$	The reference transit mid-time.
P_0	days	The observed orbital period.
P_a	days	The anomalistic orbital period.
P_s	days	The sidereal orbital period.
e_0	Dimensionless	Orbital eccentricity
ω_p	deg	The argument of periastron of the planetary orbit.
E	Dimensionless	Epoch of the observed transit.
$\frac{dP}{dE}$ or $\dot{P}$	days E^{-1}	The rate of change of orbital period with respect to the epoch. (PdE)
$\frac{d\omega}{dE}$ or $\dot{\omega}$	rad E^{-1}	The rate of change of argument of pericenter with respect to the epoch. (WdE)
a	AU	Semi-major axis of the planetary orbit.
n	rad s^{-1}	Mean motion of the planet.
I	deg	Orbital inclination relative to the stellar equator.
$M_\star$	$M_\odot$	Mass of the host star.
$R_\star$	$R_\odot$	Radius of the host star.
J_2	Dimensionless	The quadrupole moment coefficient of the host star (stellar oblateness).
$\dot{\Omega}$	rad E^{-1}	The rate of change of the longitude of the ascending node with respect to the epoch.
$\dot{M}$	rad E^{-1}	The rate of change of the mean anomaly with respect to the epoch.
T_{actual}	$\mathrm{BJD_{TDB}}$	The geometrically derived actual crossing time of a transit or eclipse.
X, Y, Z	Cartesian coordinates	The 3D relative position of the planet.
r_{proj}	projected distance	The projected distance of the planet from the star's center on the sky-plane.
$\mathcal{Z}$	Dimensionless	Bayesian evidence used for statistical model comparison.

Coordinate System

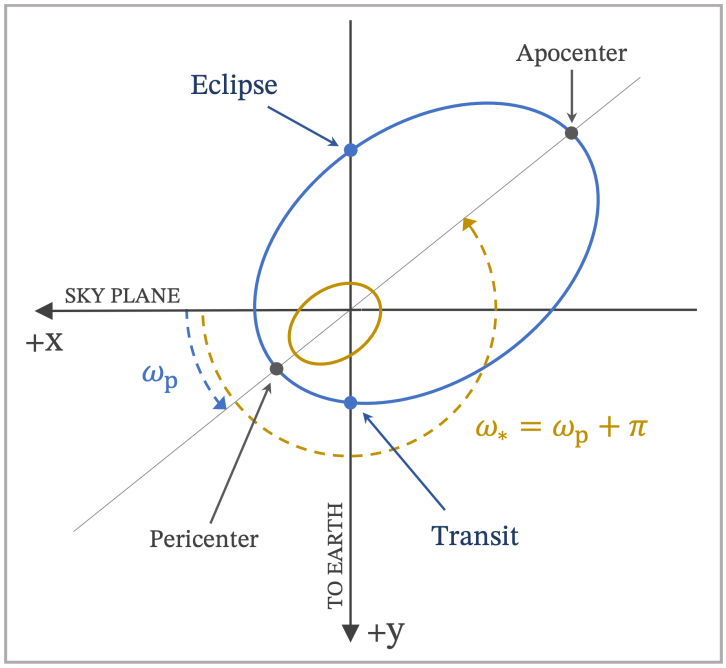


Figure 1: Coordinate system for the transit and eclipse of a planet around a star

Acknowledgements

First and foremost, I would like to express my deepest gratitude to my supervisor, Dr. Aaron Boley. I am incredibly thankful for his unwavering support, guidance, and infectious optimism throughout this work. I am especially grateful for the opportunity to explore a new and challenging research direction under his supervision. Our discussions on the complexities of WASP-12b and WASP-43b were the foundation and inspiration for this thesis.

I am also sincerely grateful to Simone Hagey, whose foundational work in secular exoplanet modeling provided the basis upon which this research was built. Her contributions made the extended analysis and implementation in this dissertation possible.

I would like to express my appreciation to my home country, Indonesia, for providing me with the opportunity through scholarship support to pursue higher education. This opportunity has allowed me to explore, question, and learn—privileges that I do not take for granted.

Finally, I would like to thank my family for their unwavering encouragement, patience, and unconditional love throughout my academic journey. Their support and sacrifices have given me the freedom to pursue my studies wholeheartedly, and this work would not have been possible without them.

Chapter 1

Introduction

1.1 Exoplanets

Exoplanets, or planets orbiting stars beyond our Solar System, have become a central focus of modern astrophysics following their first confirmed detections in the 1990s [2]. Advances in observational techniques have led to the discovery of thousands of such systems, revealing a remarkable diversity in planetary masses, orbital configurations, and dynamical behaviors. These discoveries have challenged traditional models of planetary formation and evolution, highlighting the importance of precise measurements in understanding their long-term dynamics. In particular, high-precision observations of orbital motion provide a powerful means of probing subtle physical processes that can alter planetary orbits over time. Such processes include tidal interactions, relativistic effects, and stellar structure influences, all of which can leave detectable imprints on observable quantities. As a result, exoplanet systems serve as natural laboratories for studying orbital evolution under a wide range of physical conditions.

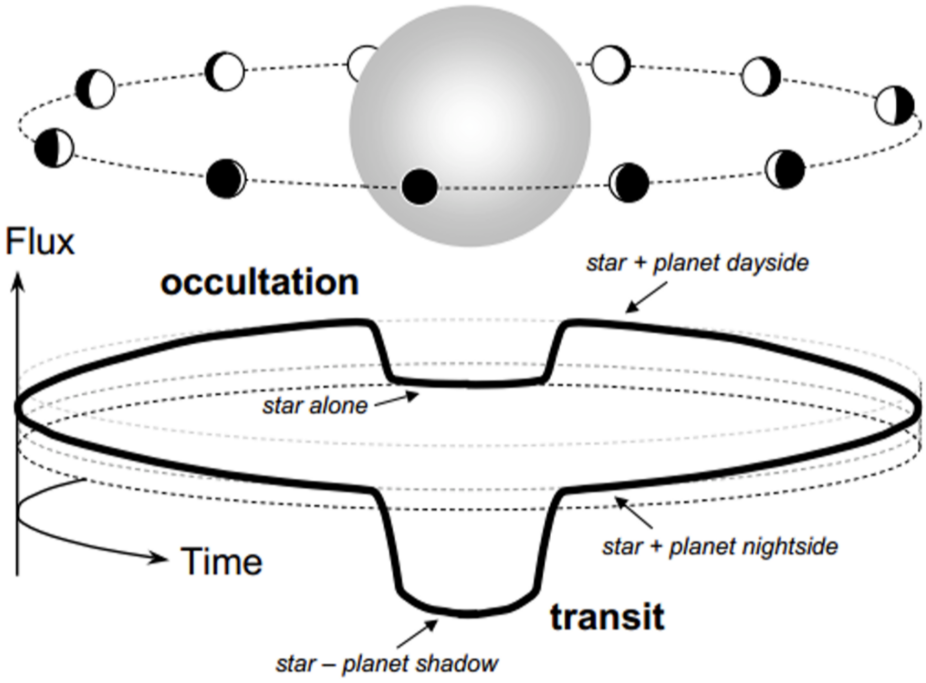


Figure 1.1: Transit and Eclipse of a planet around a star [1]

1.2 Transit Photometry

One of the most successful techniques for detecting and characterizing exoplanets is transit photometry [3–5], which measures the periodic dimming of a star as a planet passes in front of it along the line of sight(Transit), or when the planet is behind the star(Eclipse) like in figure 1.1 [6]. This method provides direct access to key orbital parameters, including the orbital period, transit duration, and relative planetary size as shown in figure 1.2. The high precision achievable with modern space-based missions [7, 8] has enabled the detection of minute variations in transit and eclipse light curves [9], allowing not only the confirmation of planetary candidates but also detailed studies of their orbital dynamics. In particular, repeated transit and eclipse observations establish a timing sequence that can be used to track deviations from strictly periodic motion. These deviations, known as Transit Timing Variations (TTVs), arise from a variety of physical processes and provide a sensitive probe of additional dynamical effects within the system [10, 11]. Worth to mention that eclipse dimming is far less apparent to our detector compared to the transit, hence the data availability is far less.

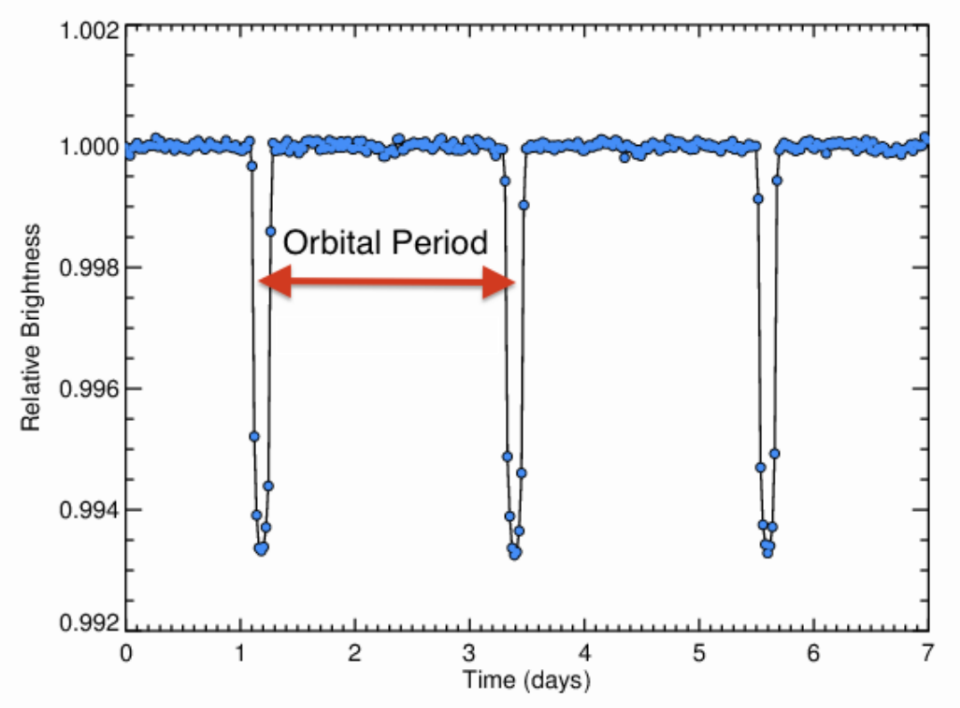


Figure 1.2: Orbital Period of a planet around a star

1.3 Transit Timing Variations (TTVs)

Transit Timing Variations (TTVs) refer to deviations in the observed times of planetary transits from a strictly periodic sequence. In an ideal two-body system, transits occur at regular intervals determined by a constant orbital period. However, in realistic systems, a variety of physical processes can introduce small but measurable shifts in these timings. These include gravitational interactions with additional bodies, tidal dissipation, and relativistic or stellar-induced perturbations.

TTVs provide a powerful diagnostic tool for probing such effects, as even small perturbations can accumulate over time and produce detectable signatures in transit data. Depending on the underlying mechanism, TTV signals may exhibit distinct trends, such as monotonic drifts or periodic modulations, reflecting changes in the orbital period or orientation [12, 13]. As a result, TTV analysis has become an important method for investigating the dynamical evolution of exoplanetary systems.

In many studies, these variations are interpreted under the assumption that a single dominant mechanism governs the observed signal. However, multiple processes may operate simultaneously, potentially leading to more complex TTV signatures that cannot be fully captured by single-effect models. This work investigates whether neglecting such simultaneous effects can lead to systematically biased interpretations.

1.4 Transit and Eclipse Timing Model

The interpretation of Transit Timing Variations (TTVs) relies on constructing models that describe how the expected transit times evolve over successive orbital cycles. In the simplest case, transits occur periodically with a constant orbital period. However, physical processes such as tidal dissipation and orbital precession can introduce systematic deviations from this periodic behavior. In this section, we outline three commonly used timing models: the constant-period model, orbital decay model, and precession model. To aid the understanding of the specific geometric and mathematical formulas presented here, a full list of relevant variable definitions along with an illustrated reference coordinate system (Figure 1) are provided in the Nomenclature section at the beginning of this thesis.

1.4.1 Constant Period Model

In the absence of perturbations, the orbit of a planet can be approximated as strictly periodic, with a constant orbital period. The expected mid-transit and mid-eclipse time at epoch E is given by [5]

$$T(E) = T_0 + P_0 E \tag{1.1}$$

$$T(E) = T_0 + P_0 E + \frac{1}{2} P_0 + \frac{2}{\pi} P_0 e \cos(\omega_p) \tag{1.2}$$

with Eq. 1.1 is for transit and Eq. 1.2 is for eclipse. This linear calculation serves as the baseline model against which deviations are measured. Any systematic departure from this relation indicates the presence of additional physical effects influencing the orbital dynamics which we will plot as an $O - C$ (Occultation minus Calculated) diagram.

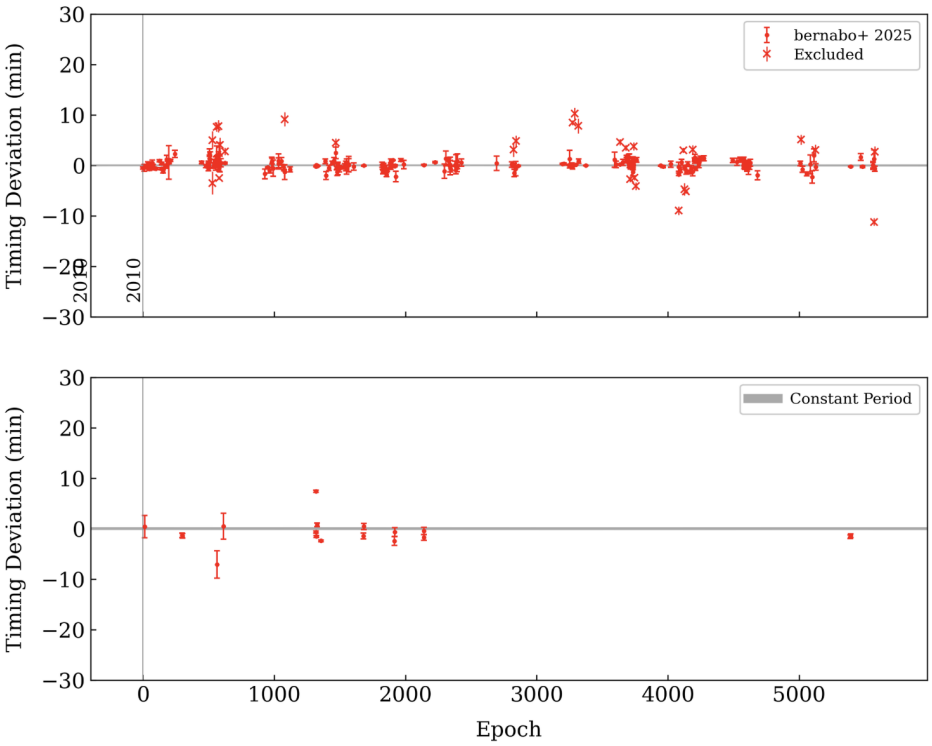


Figure 1.3: a) Transit and b) Eclipse timing of WASP 43b using constant period model

One notable example is WASP-43b, a hot Jupiter orbiting a K-type star, which has been extensively monitored using transit and eclipse timing observations (Figure 1.3). Despite its extremely short orbital period, making it highly sensitive to tidal effects, recent high-precision analyses have found that its transit times remain consistent with a constant-period (linear) model within observational uncertainties. This makes WASP-43b an important benchmark system for validating timing methodologies and assessing the detectability of small deviations from strict periodicity [14]. The absence of significant departures from a constant-period model also places constraints on the efficiency of tidal dissipation and the magnitude of possible dynamical perturbations in the system.

1.4.2 Decay Model

Orbital decay arises from dissipative processes, such as tidal interactions between the planet and its host star, leading to a gradual decrease in the orbital period over time. In this case, the period is no longer constant but evolves with epoch. To first order, this can be modeled by introducing a quadratic term in the transit timing relation [15, 16]:

$$T(E) = T_0 + P_0 E + \frac{1}{2}\dot{P}E^2 \tag{1.3}$$

$$T(E) = T_0 + P_0 E + \frac{1}{2}P_0 + \frac{1}{2}\dot{P}E^2 + \frac{2}{\pi}P_0 e\cos(\omega_p) \tag{1.4}$$

with Eq. 1.3 is for transit and Eq. 1.4 is for eclipse. This quadratic calculation produces a quadratic deviation from the linear model, corresponding to a cumulative shift in transit times.

The sign and magnitude of $\dot{P}$ determine whether the orbit is decaying or expanding, although in most close-in exoplanet systems, tidal effects are expected to drive orbital decay [15–17].

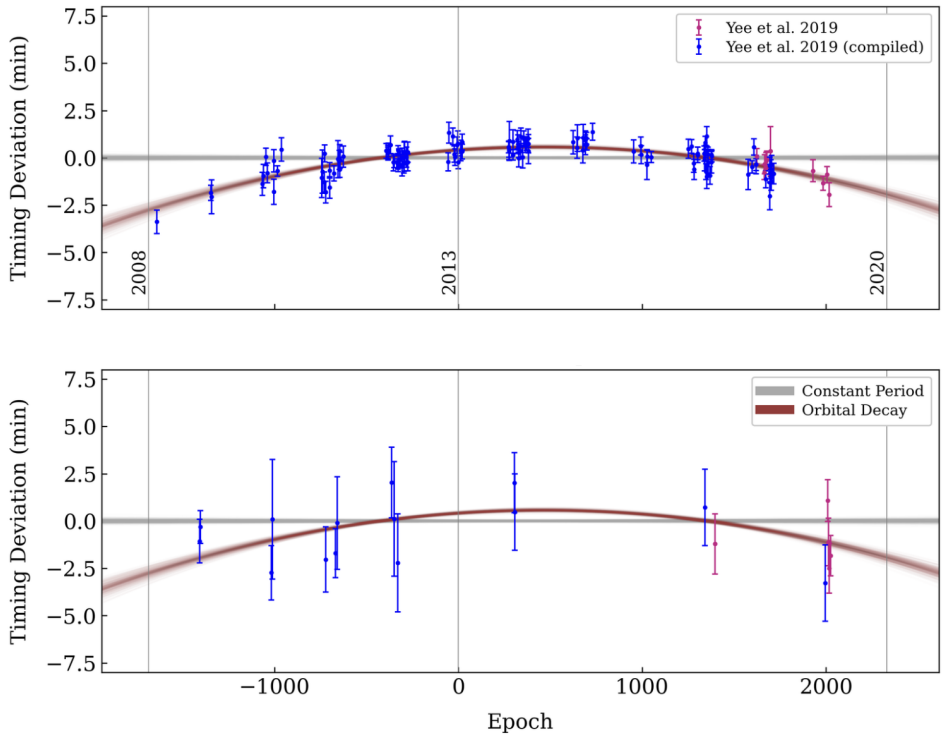


Figure 1.4: a) Transit and b) Eclipse timing of WASP 12b using decay model

The hot Jupiter WASP-12b provides one of the clearest observational examples of deviations from a standard constant-period model (Figure 1.4). High-precision transit timing measurements have revealed a systematic shift in transit times consistent with a decreasing orbital period, indicative of tidal orbital decay. Detailed analyses, including those by Yee et al. [18], demonstrate that the timing data are better described by a quadratic decay model rather than a fixed linear one. This behavior is widely interpreted as evidence of efficient tidal dissipation within the host star, leading to the gradual inspiral of the planet. As such, WASP-12b serves as a benchmark system for studying tidal evolution and constraining stellar tidal quality factors.

1.4.3 Precession Model

Orbital precession refers to the gradual rotation of the orbit within its plane, which can arise from general relativistic effects, tidal distortions, or stellar oblateness [19–21]. Unlike orbital decay, precession does not primarily change the orbital period but instead alters the orientation of the orbit, leading to periodic variations in the observed transit times.

For an eccentric orbit, the precession of the argument of periastron modifies the timing of transits relative to a fixed reference frame. This results in a quasi-periodic TTV signal that depends on the eccentricity and precession rate. To leading order, the timing variation can be expressed as a periodic modulation superimposed on the baseline linear model [19, 20]:

$$T(E) = T_0 + P_s E - \frac{1}{\pi} P_0 e \cos(\omega_p) \tag{1.5}$$

$$T(E) = T_0 + P_s E + \frac{1}{2} P_a + \frac{1}{\pi} P_0 e \cos(\omega_p) \tag{1.6}$$

with Eq. 1.5 is for transit and Eq. 1.6 is for eclipse. where here ω_p changes with time as

$$\omega_p(E) = \omega_0 + \dot{\omega}_p E \tag{1.7}$$

This model produces oscillatory timing deviations, distinguishing it from the quadratic trend associated with orbital decay. Worth to mention that the sign of the $\cos(\omega_p)$ term is opposite for transit and eclipse, which means that the TTVs for transit and eclipse will be out of phase. It also help us later to distinguish between the precession model and the decay model.

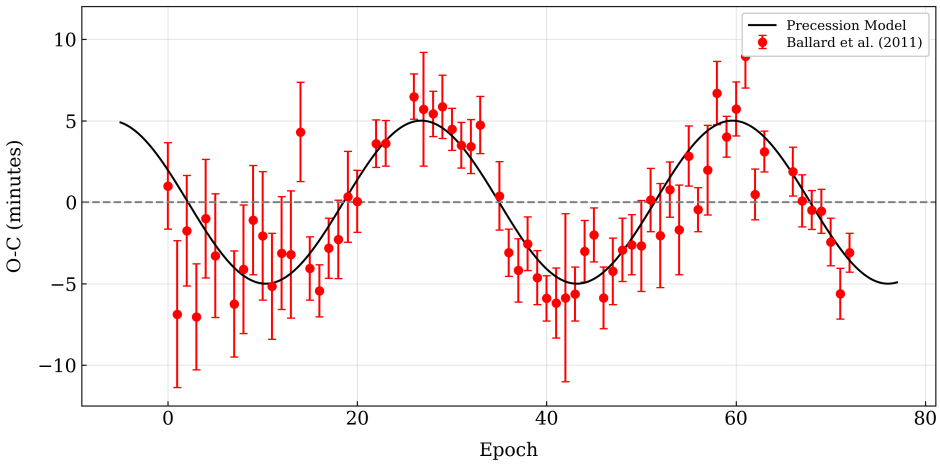


Figure 1.5: Transit timing of Kepler-19b using precession model

A clear observational demonstration of transit timing variations is provided by the Kepler-19 system (Figure 1.5). Transit observations of Kepler-19b reveal timing deviations with amplitudes of several minutes that cannot be explained by a simple constant-period model. These variations are instead attributed to gravitational perturbations from an additional, non-transiting planetary companion, making this one of the earliest and most compelling detections of unseen planets via the TTV method [22].

Chapter 2

Thesis

2.1 Motivation: Simultaneous Effect

As discussed previously, orbital decay is driven by tidal dissipation within the star–planet system, leading to a gradual loss of orbital energy and a secular decrease in the orbital period. In contrast, apsidal precession arises from a combination of effects, including tidal deformation, stellar quadrupole moments, perturbations from additional companions, and general relativistic corrections, resulting in a time-dependent rotation of the argument of periastron.

Given that these mechanisms originate from fundamentally different physical processes, there is no a priori reason to assume that they are mutually exclusive. In principle, both effects may act simultaneously, producing a composite signal in transit timing observations. Nevertheless, most existing studies treat orbital decay and apsidal precession independently, largely due to the degeneracy between their observational signatures and the limited temporal baseline of available data.

This issue is exemplified in the case of WASP-12b, where observed transit timing variations have been interpreted as evidence for orbital decay, while alternative analyses have shown that apsidal precession can reproduce similar trends. This highlights the difficulty in uniquely identifying the underlying physical mechanism from timing data alone.

In this thesis, we address this limitation by constructing a model that incorporates both orbital decay and apsidal precession simultaneously. We explore how the interplay between these effects influences the resulting transit timing variations and assess the extent to which they can be disentangled through statistical modeling.

2.2 Brief Model Recap

Before constructing a comprehensive framework, it is instructive to briefly recap the independent analytical forms for each transit timing model introduced in the previous chapter. The predicted time of a transit event, $T(E)$, at an epoch E can be modeled depending on the dominant physical perturbation:

1. **Constant-Period Model:** Assuming a strictly periodic Keplerian orbit with no external perturbations, the transit times follow a linear baseline. The expected transit and eclipse times are given by:

$$T(E)_{tra} = T_0 + P_0 E \tag{2.1}$$

$$T(E)_{ecl} = T_0 + P_0 E + \frac{1}{2} P_0 + \frac{2}{\pi} P_0 e \cos(\omega_p) \tag{2.2}$$

   where T_0 is the reference epoch, P_0 is the constant orbital period, e is the orbital eccentricity, and ω_p is the argument of periastron.

2. **Orbital Decay Model:** When tidal dissipation continuously drains orbital energy, the period decreases secularly over time. This leads to a quadratic deviation in the timing

formulas:

$$T(E)_{tra} = T_0 + P_0 E + \frac{1}{2}\dot{P}E^2 \tag{2.3}$$

$$T(E)_{ecl} = T_0 + P_0 E + \frac{1}{2}P_0 + \frac{1}{2}\dot{P}E^2 + \frac{2}{\pi}P_0 e\cos(\omega_p) \tag{2.4}$$

where $\dot{P}$ represents the decay rate per epoch.

3. **Apsidal Precession Model:** In eccentric orbits subject to apsidal precession, the argument of periastron rotates over time ($\omega_p(E) = \omega_0 + \dot{\omega}_p E$). This induces a periodic sinusoidal modulation on the timing:

$$T(E)_{tra} = T_0 + P_s E - \frac{1}{\pi}P_0 e\cos(\omega_p) \tag{2.5}$$

$$T(E)_{ecl} = T_0 + P_s E + \frac{1}{2}P_a + \frac{1}{\pi}P_0 e\cos(\omega_p) \tag{2.6}$$

where P_s is the sidereal period and P_a is the anomalistic period.

These independent analytical expressions serve as the foundation. However, to account for environments where both tidal dissipation and orbital precession actively compete, these elements must be synthesized.

2.3 Combined Decay-Precession Model

Understanding whether orbital decay, apsidal precession, or a combination of both governs the observed transit timing variations is crucial for correctly interpreting the underlying physics of exoplanet systems. Each mechanism encodes different information: orbital decay is directly related to tidal dissipation efficiency and therefore constrains the internal structure and tidal quality factors of the star and planet, while apsidal precession provides insight into orbital geometry, planetary interior structure, and additional dynamical influences such as unseen companions or relativistic effects. Misattributing one effect for the other can therefore lead to incorrect conclusions about the physical properties and long-term evolution of the system. By considering both processes simultaneously, this work aims to reduce such ambiguities and provide a more comprehensive framework for interpreting transit timing data, ultimately improving our ability to extract reliable physical parameters from high-precision observations.

In this work, we assume that orbital decay and apsidal precession arise from independent physical processes and can therefore be treated as separable contributions to the transit timing signal. Under this assumption, we construct a combined model, referred to as the Decay–Precession (DP) model, which incorporates both effects simultaneously. The transit and eclipse timing equations are given by:

$$T(E)_{\mathrm{tra}} = T_0 + P_0 E + \frac{1}{2}\dot{P}E^2 - \frac{P_0}{\pi}e\cos\big(\omega(E)\big) \tag{2.7}$$

$$T(E)_{\mathrm{ecl}} = T_0 + P_0 E + \frac{1}{2}P_0 + \frac{1}{2}\dot{P}E^2 + \frac{P_0}{\pi}e\cos\big(\omega(E)\big) \tag{2.8}$$

where the argument of periastron evolves as

$$\omega(E) = \omega_0 + \dot{\omega}E \tag{2.9}$$

The DP model therefore captures both the secular evolution of the orbital period and the periodic modulation due to apsidal motion within a unified framework.

2.4 Simulation Framework

Having established the analytical timing models, we now construct a simulation framework to generate synthetic data and evaluate their observable signatures.

A controlled simulation environment is necessary because the simultaneous occurrence of orbital decay and apsidal precession has not yet been conclusively observed or systematically modeled in any known exoplanet system. As a result, there is no observational dataset with a known ground truth that can be used to validate the combined Decay-Precession (DP) model. Simulation therefore provides a controlled setting in which the true physical parameters are known, allowing for a direct assessment of model performance and parameter recovery.

Within this framework, orbital dynamics are mapped onto observable transit timing variations (TTVs). For a given set of input parameters, the model generates synthetic transit and eclipse times at discrete observational epochs under different physical scenarios: constant period, orbital decay, apsidal precession, and their combined effects. These timing sequences are then converted into observed-minus-calculated (O–C) signals, which serve as the primary diagnostic for comparison with fitted models.

Because the parameter space associated with combined perturbative effects can be highly correlated, the framework is designed to efficiently evaluate timing models over long temporal baselines spanning hundreds to thousands of orbital cycles. This enables a systematic exploration of regimes in which different physical processes dominate, overlap, or become observationally degenerate.

To generate these expansive simulated datasets without encountering prohibitive computational costs, the simulation engine relies on a highly efficient discrete orbital algorithm. In the following subsection, we detail the core geometric and mathematical structure of this stroboscopic forward-modeling approach.

2.4.1 Stroboscopic Approach for Precession and Decay

Direct numerical integration of orbital motion provides a complete description of the system but is computationally expensive and generally unnecessary for modeling transit timing variations, which depend only on the orbital configuration at specific phases. To address this, we adopt a *stroboscopic approach*, in which the orbital state is evaluated discretely at the reference epochs corresponding to primary transits and secondary eclipses. Similar phase-sampling concepts have been widely used in transit timing analyses to efficiently model timing variations without requiring full continuous integration [10, 23]. This approach is particularly well-suited for parameter inference, as it directly computes observable quantities without introducing unnecessary computational overhead from unobserved orbital phases.

The underlying formulation and codebase for the geometric stroboscopic algorithm was developed by Dr. Aaron Boley for modeling apsidal precession. In this work, we extend this framework by incorporating a variable mean motion to simulate secular orbital decay ($\dot{P}$), enabling the simultaneous treatment of both effects.

The geometric configuration adopted in this work is illustrated in Figure 2.1, adapted from [24]. This coordinate system provides a consistent framework for relating orbital elements to the observer's line of sight and for defining the transit condition through projected sky-plane distances.

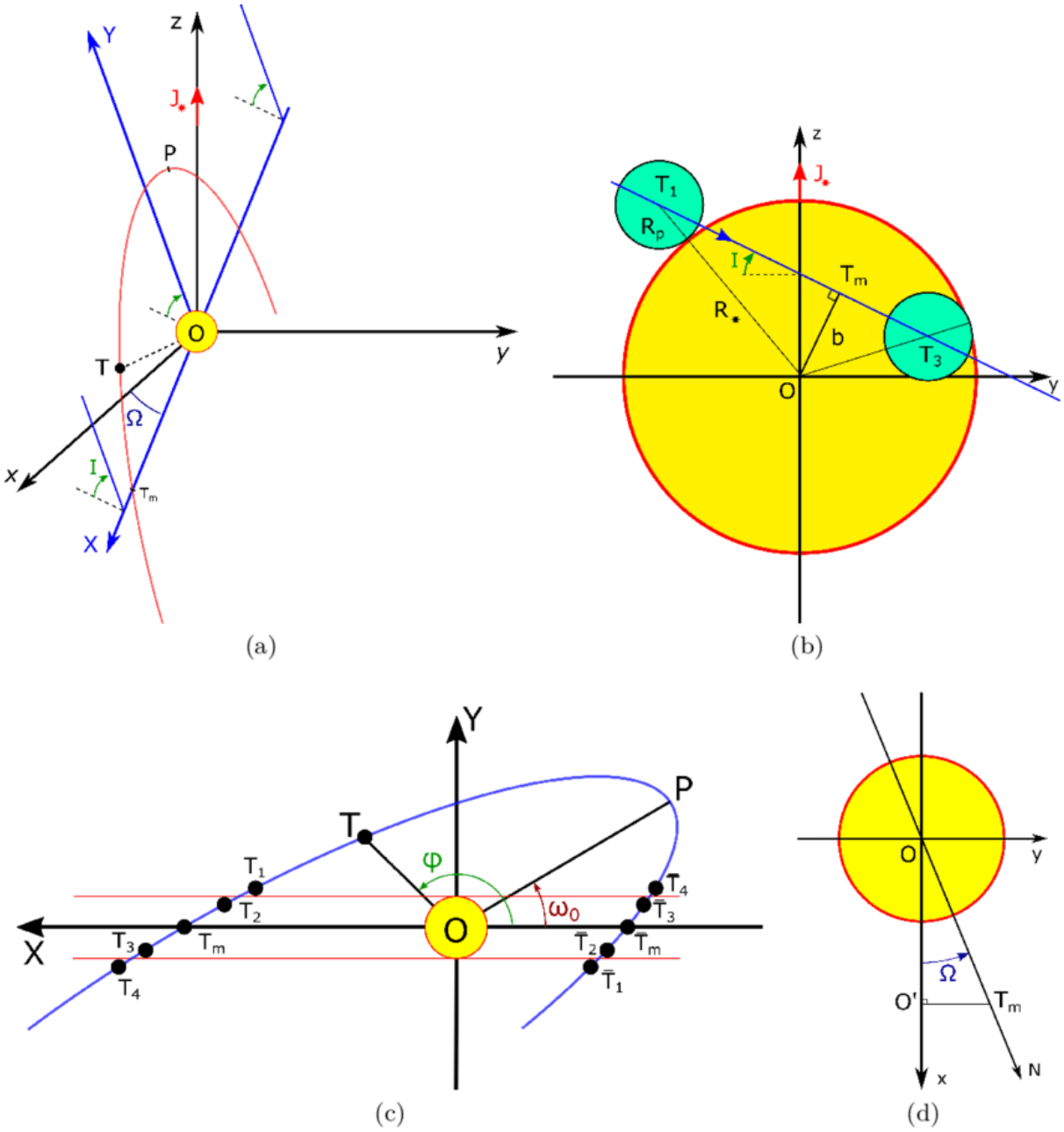


Figure 2.1: Geometry of the planetary transit adapted from [24]. (a) Three-dimensional configuration of the system in the observer frame (x, y, z), where the x-axis is aligned with the line of sight and the (y, z) plane defines the sky plane. The stellar center is denoted by O, and the orbital periastron is labeled P. The stellar spin axis is projected onto the sky plane. (b) Projection of the transit chord as seen by the observer, illustrating the definition of contact points. (c) Orbital-plane view (X, Y) showing the planet trajectory and the locations of transit (T_i) and eclipse $(\bar{T}_i)$ events. (d) Top-down view of the orbit highlighting the line of nodes and its relation to the observer direction.

In this approach, rather than integrating the planet's continuous trajectory, the system is advanced directly to a target epoch E (or $E+\frac{1}{2}$ for eclipses). First, the apsidal precession is natively implemented through a parameterized secular advance of the argument of periastron $(\dot{\omega})$.

For physical interpretation, one representative mechanism considered here is the rotationally induced oblateness of the host star, which introduces a quadrupole moment in the gravitational potential parameterized by the coefficient J_2. This serves as a concrete example of how precession can arise from stellar structure, although the framework itself remains general and can accommodate other sources of precession. In this formulation, the perturbation induces secular evolution of the orbital elements according to classical mechanics:

$$\dot{\omega} = \frac{3}{4} n J_2 \left(\frac{R_\star}{a}\right)^2 \frac{5\cos^2 I - 1}{(1-e^2)^2}, \tag{2.10}$$

$$\dot{\Omega} = -\frac{3}{2} n J_2 \left(\frac{R_\star}{a}\right)^2 \frac{\cos I}{(1-e^2)^2}, \tag{2.11}$$

$$\dot{M} = n + \frac{3}{4} n J_2 \left(\frac{R_\star}{a}\right)^2 \frac{3\cos^2 I - 1}{(1-e^2)^{3/2}}, \tag{2.12}$$

where $n = \sqrt{GM_\star/a^3}$ is the mean motion, $R_\star$ is the stellar radius, a is the semi-major axis, and e is the orbital eccentricity.

To extend this framework and account for additional orbital decay, we linearly step a variable mean motion parameter across time. Specifically, we calculate the instantaneously shrinking orbital period $P(E) = P_0 + \dot{P}E$ and update the corresponding temporal reference for each specific target epoch. Based on this evolving period, the instantaneous semi-major axis and mean motion are subsequently recalculated.

Because the semi-major axis strictly decreases as the orbit shrinks under decay, the magnitude of the geometric precession terms (which scale intricately with semi-major axis, for instance a^{-2} and $a^{-7/2}$) is continuously modified. By explicitly recalculating both the decay elements and the geometry-dependent precession rates at every targeted epoch, the simulation seamlessly incorporates the two mechanisms and captures the leading-order physical coupling between orbital decay and apsidal precession.

Using these local orbital elements, the algorithm computes the eccentric and true anomalies and maps the planet's position into the observer frame (X, Y, Z), with the X-axis aligned along the line of sight. As illustrated in Figure 2.1, the mid-transit (or mid-eclipse) time corresponds to the minimum projected sky-plane distance,

$$r_{\rm proj} = \sqrt{Y^2 + Z^2}, \tag{2.13}$$

which is equivalent to solving

$$\frac{d}{dt}(Y^2 + Z^2) = 2Y\frac{dY}{dt} + 2Z\frac{dZ}{dt} = 0. \tag{2.14}$$

To determine this crossing efficiently, the algorithm evaluates the system at two nearby time points and applies a secant root-finding method to interpolate the precise moment at which the derivative vanishes. The resulting transit timing variation is then defined relative to a constant-period ephemeris,

$$(O - C) = T_{\rm actual} - (T_0 + P_0 E). \tag{2.15}$$

By advancing the orbital elements discretely and restricting geometric calculations to the immediate vicinity of the transit or eclipse, this method avoids full continuous integration while maintaining high accuracy. The resulting synthetic O–C signals capture both secular and periodic timing variations with significantly reduced computational cost, making the approach well-suited for large-scale Bayesian inference. The exact algorithmic pseudocode detailing this stroboscopic generation process is provided in Appendix A.

2.4.2 Simulation for Different Models

To illustrate the qualitative differences between the timing models, we generate synthetic transit timing variations using the simulation framework described above. Because the simulation is built on explicit physical parameters, we can isolate each mechanism by selectively enabling or disabling the relevant effects.

The *orbital decay* model is simulated by introducing a non-zero secular period change ($\dot{P} \neq 0$) while assuming a spherically symmetric stellar potential ($J_2 = 0$), thereby eliminating precession. Conversely, the *apsidal precession* model is generated by setting $J_2 \neq 0$ for an eccentric orbit ($e \neq 0$) while keeping the orbital period constant ($\dot{P} = 0$). The combined *Decay-Precession (DP)* model is constructed by activating both effects simultaneously.

Figure 2.2 presents the resulting O–C diagrams for these scenarios. The orbital decay model exhibits a characteristic quadratic trend due to the continuous secular decrease in orbital period. The precession model produces periodic oscillations arising from the rotation of the argument of periastron. The combined DP model displays a superposition of these effects, yielding an oscillatory signal modulated by a long-term quadratic drift.

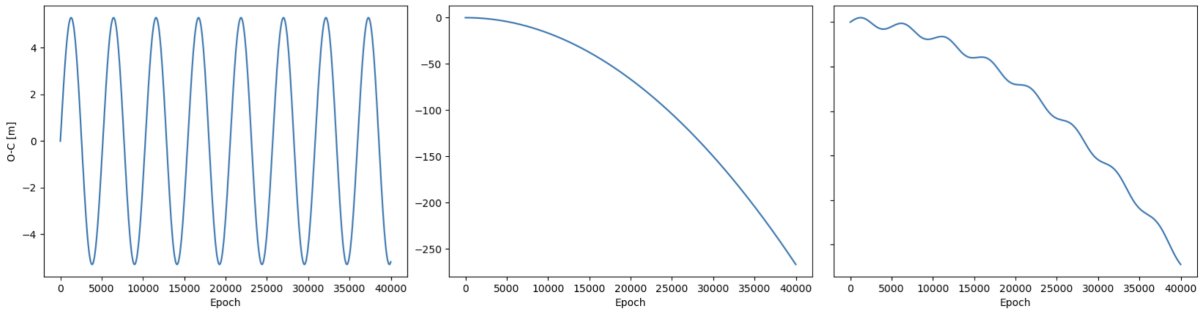


Figure 2.2: Simulated O–C diagrams illustrating the qualitative behavior of different timing models: (a) apsidal precession, showing periodic oscillations; (b) orbital decay, producing a quadratic secular trend; and (c) the combined Decay-Precession (DP) model, exhibiting a superposition of periodic modulation and long-term drift.

2.5 Fitting Method

To assess whether these models can be distinguished observationally, we now introduce a statistical framework for parameter estimation and model comparison. Specifically, to rigorously analyze the transit timing data and infer the underlying orbital parameters, we adopt a Bayesian inference framework implemented via nested sampling. While traditional least-squares methods are effective for identifying best-fit solutions, they are often limited in their ability to explore complex or degenerate parameter spaces and do not provide a natural measure for model comparison. In contrast, nested sampling enables a global exploration of the parameter space while simultaneously computing the Bayesian evidence ($\mathcal{Z}$), which is essential for statistically comparing competing models.

This approach allows us to quantitatively evaluate which physical model—namely the Constant-Period, Orbital Decay, Apsidal Precession, or the combined Decay-Precession (DP) model—provides the most consistent description of the observed transit timing variations.

The computational implementation is based on the `orbdot` package [25, 26], an open-source Python framework designed to model the secular evolution of exoplanetary orbits. In this

work, we extend the functionality of `orbdot` by incorporating the analytical Decay-Precession (DP) model directly into its evaluation pipeline. This modification enables the simultaneous modeling of both secular orbital decay and periodic apsidal precession within a unified Bayesian framework.

2.5.1 Nested Sampling Algorithm

Nested sampling is a Monte Carlo technique introduced by Skilling [27] to efficiently compute the Bayesian evidence while also generating posterior distributions for model parameters. Unlike Markov Chain Monte Carlo (MCMC) methods, which are primarily designed to sample posterior distributions and may struggle with multimodal or highly degenerate likelihood surfaces, nested sampling is explicitly constructed to evaluate the evidence integral over the full prior volume.

The algorithm operates by maintaining a set of "live points" that are initially distributed according to the prior. At each iteration, the point with the lowest likelihood is removed and replaced by a new sample drawn from the prior under the constraint that its likelihood exceeds that of the discarded point. This iterative process progressively restricts the sampling region to nested shells of increasing likelihood, effectively concentrating computational effort on the most relevant regions of parameter space.

The outcome of this process is a set of posterior samples that encode both the most probable parameter values and their correlations. This is illustrated in Figure 2.3, which shows a corner plot derived from the nested sampling analysis of the orbital decay model. The diagonal panels represent the marginalized posterior distributions for each parameter, while the off-diagonal panels show the joint distributions, revealing correlations between parameters.

In particular, the correlations visible in the corner plot indicate that certain parameters are not independent, as variations in one parameter can be partially compensated by adjustments in another. For example, the correlation between P_0 and $\dot{P}$ reflects the fact that small changes in the orbital period can mimic long-term trends over a limited observational baseline.

However, the well-localized and unimodal nature of the posterior distributions demonstrates that the parameters remain well-constrained, and no strong degeneracies are present in the model.

2.6 Model Comparison via Bayesian Evidence

To determine which physical model best describes the observed transit timing variations, we perform model selection using the Bayesian evidence ($\mathcal{Z}$) computed through nested sampling [27].

The Bayesian evidence is defined as:

$$\mathcal{Z} = \int \mathcal{L}(\theta)\, \pi(\theta)\, d\theta, \tag{2.16}$$

where $\mathcal{L}(\theta)$ is the likelihood function and $\pi(\theta)$ represents the prior distribution over the parameter space θ. The evidence naturally incorporates both the goodness-of-fit and the complexity of the model, penalizing models with excessive parameter freedom [28].

To compare two competing models M_i and M_j, we compute the Bayes factor:

$$B_{ij} = \frac{\mathcal{Z}_i}{\mathcal{Z}_j}, \tag{2.17}$$

or equivalently in logarithmic form:

$$\Delta \ln \mathcal{Z} = \ln \mathcal{Z}_i - \ln \mathcal{Z}_j. \tag{2.18}$$

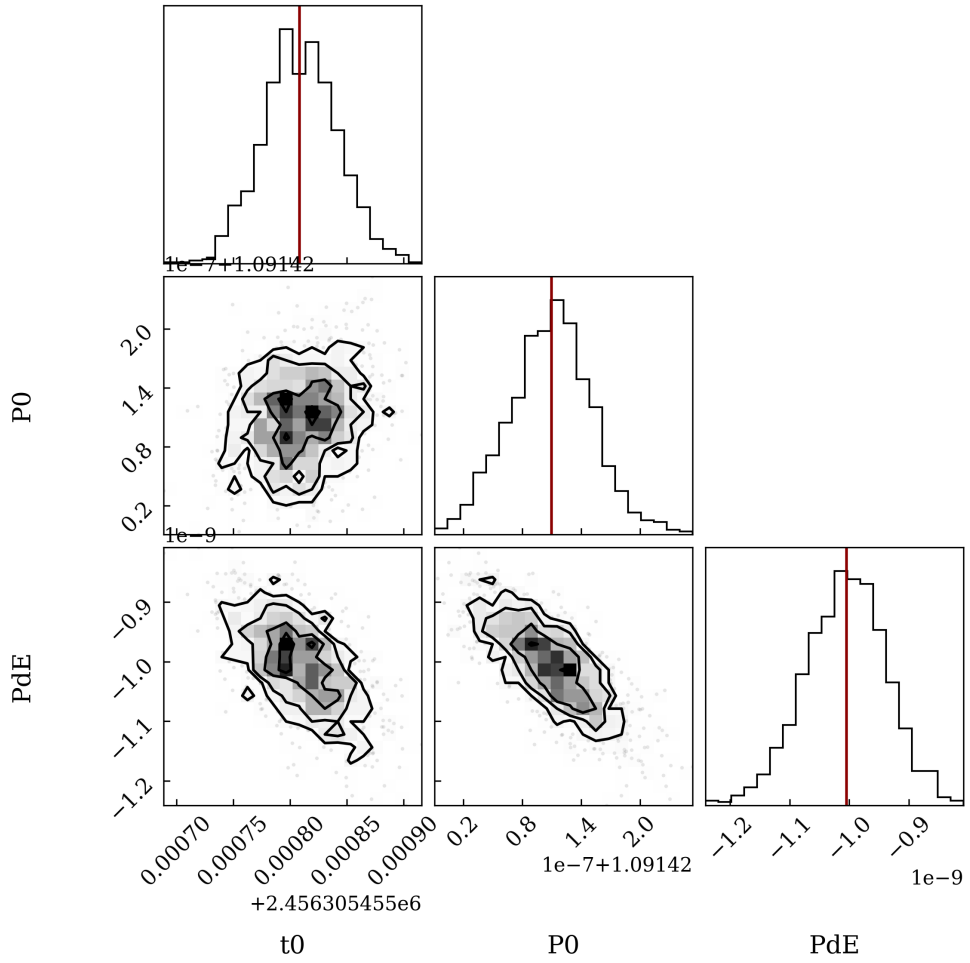


Figure 2.3: Corner plot of posterior distributions obtained from nested sampling for the orbital decay model. The diagonal panels show the marginalized distributions of each parameter, while the off-diagonal panels display joint distributions, highlighting correlations such as those between T_0, P_0, and $\dot{P}$.

The significance of model preference is interpreted using a modified Jeffreys scale, based on the difference in log Bayesian evidence ($\Delta \ln \mathcal{Z}$), as summarized in Table 2.1 [28].

Table 2.1: Interpretation of differences in log Bayesian evidence ($\Delta \ln \mathcal{Z}$) for model comparison.

$\Delta \ln \mathcal{Z}$	Strength of Evidence
< 1	Inconclusive
$1 - 2.5$	Weak evidence
$2.5 - 5$	Moderate evidence
> 5	Strong evidence

In this work, $\Delta \ln \mathcal{Z}$ is computed relative to the constant-period model, which serves as the

baseline hypothesis. The model with the highest evidence is statistically preferred, with the magnitude of $\Delta \ln \mathcal{Z}$ indicating the strength of the preference.

This framework provides a systematic and quantitative method for assessing whether the observed timing variations are best explained by a single physical mechanism or by a combination of orbital decay and apsidal precession.

Chapter 3

Results

3.1 Overview of Simulation Study

We now present the results of applying this framework to the simulated datasets. Because the simultaneous presence of orbital decay and apsidal precession has not yet been observationally established or systematically modeled in transit timing analyses, there is currently no well-constrained real exoplanet dataset that can be used to directly test such a combined fitting framework. As a result, we adopt a simulation-based approach, which allows us to generate controlled datasets that mimic realistic observations while retaining full knowledge of the underlying physical parameters. This provides a critical advantage: the true values of the system are known a priori, enabling a direct and quantitative assessment of both model accuracy and parameter recovery.

To investigate the scenario in which orbital decay and apsidal precession occur simultaneously, we construct synthetic datasets using the simulation framework described in Section 2.4 and analyze them with the fitting methodology outlined in Section 2.5. This approach enables a controlled comparison between the true underlying physical parameters and those recovered by each model.

The workflow of the simulation study is summarized as follows. First, synthetic transit and eclipse timing data are generated for each combination of parameters listed in Table 3.1, ensuring comprehensive coverage of the relevant parameter space. The exact algorithmic logic used to systematically sweep through these grid combinations, inject noise, and aggressively automate the nested sampling fits is detailed in Appendix B.

To emulate realistic observations, Gaussian noise is added to the simulated timing data, with a standard deviation chosen to reflect current observational precision. This step ensures that the datasets are representative of real transit timing measurements.

The resulting datasets are then fitted using four candidate models: constant period, orbital decay, apsidal precession, and the combined Decay-Precession (DP) model. The fitting is performed within a Bayesian framework using nested sampling, which yields both posterior parameter distributions and Bayesian evidence values for each model.

Finally, model performance is evaluated through two complementary metrics: Bayesian evidence, which determines the statistically preferred model, and parameter recovery accuracy, assessed by comparison with the known input values. This dual evaluation allows us to quantify both the statistical and physical reliability of each model across different regimes of the parameter space.

3.2 Case Studies

To systematically evaluate the performance of the proposed models, we generate synthetic datasets over the full combinatorial parameter space defined in Table 3.1. Specifically, we consider all possible combinations of the number of data points, orbital eccentricity, orbital

decay rate ($\dot{P}$), and apsidal precession rate ($\dot{\omega}$), resulting in a total of $3 \times 6 \times 6 \times 6 = 648$ simulated cases.

For each parameter combination, we generate synthetic transit timing data and assess whether the combined Decay-Precession (DP) model provides a statistically improved description compared to the individual models. The complete set of results generated from these 648 simulated grid combinations—including their associated parameter traces and Bayesian evidence extractions—is publicly available in the accompanying repository at https://github.com/VavanZaa/combined-transit-model-simulations.

Table 3.1: Parameter ranges explored in the simulation.

Number of Data Points	Eccentricity (e)	$\dot{P}$	$\dot{\omega}$
1000	0	0	0
2000	10^{-5}	10^{-12}	10^{-8}
3000	10^{-4}	10^{-11}	10^{-7}
	10^{-3}	10^{-10}	10^{-6}
	10^{-2}	10^{-9}	10^{-5}
	10^{-1}	10^{-8}	10^{-4}

The selected parameter ranges are chosen to reflect values commonly inferred in exoplanetary systems, spanning several orders of magnitude in eccentricity, orbital decay rates, and apsidal precession rates. This allows us to explore regimes in which each effect is negligible, dominant, or comparable in magnitude.

For all simulations, the initial orbital period evaluates to $P_0 \approx 4.083$ days. Rather than being an arbitrarily assigned constant, this value naturally arises directly from the fundamental physical parameters configured within the simulation architecture. Specifically, the base simulated system assumes an initial semi-major axis of $a = 0.05\,\mathrm{AU}$ around a solar-type host star with a mass of $M_\star = 1.0\,M_\odot$ and a radius of $R_\star = 1.0\,R_\odot$. Through Kepler's Third Law and the modeled J_2 orbital perturbations, these structural dimensions dictate the resulting approximately four-day characteristic timescale. While the exact absolute value of the period does not affect the relative model statistical comparisons, deriving it directly from consistent physical invariants ensures that the simulated signals remain dynamically rigorous and highly representative of typical hot Jupiter systems.

The simulated observations are sampled at discrete intervals corresponding to every 10 orbital epochs. This results in a temporal spacing of approximately 40 days between consecutive data points. Consequently, the total observational baseline scales as $10 \times N$ orbital periods, where N is the number of data points.

By varying both the physical parameters and the number of data points, this framework enables a systematic investigation of how observational coverage and signal strength influence the ability to distinguish between orbital decay, apsidal precession, and their combined effects.

3.3 Model Comparison Results

When only a single physical effect is present, the corresponding model reliably recovers the true input parameters, demonstrating that the fitting framework is well-calibrated in the absence of competing signals.

When both orbital decay and apsidal precession are present, the preferred model depends on the relative strength of the decay signal. For datasets with 1000 data points, at lower decay

rates ($\dot{P} \lesssim 10^{-10}$ days/epoch), the decay-induced secular trend is too small to be distinguished from noise or remains subdominant compared to the periodic variations. In this regime, the precession model is typically favored, as it captures the dominant observable signal.

As the decay rate increases ($\dot{P} \gtrsim 10^{-10}$ days/epoch), the secular drift becomes more pronounced and can no longer be neglected. In this regime, the combined Decay-Precession (DP) model generally provides a better description of the data, as it accounts for both the long-term trend and the periodic modulation. However, for 1000 data points, the improvement in Bayesian evidence is often not sufficient to constitute strong statistical preference, indicating that the available data are still limited in their ability to fully disentangle the two effects.

A representative subset of the fitting results is shown in Figure 3.1, illustrating the recovered parameters and Bayesian evidence values for a fixed eccentricity case with 1000 data points. As seen in the table, for lower values of $\dot{P}$, the precession model typically yields higher Bayesian evidence, despite deviations in the recovered decay parameter. As $\dot{P}$ increases, the DP model provides improved parameter recovery and becomes increasingly competitive in evidence, although the statistical preference remains limited for this dataset size.

This limitation is overcome with larger datasets. For larger datasets (2000 and 3000 data points), the threshold decay rate at which the DP model becomes preferred shifts to lower values, and the statistical evidence in favor of the DP model becomes stronger. This reflects the improved ability of longer observational baselines to resolve small secular trends.

In addition to model selection based on Bayesian evidence, we also evaluate the accuracy of the recovered parameters by comparing them to the true input values used in the simulations. Interestingly, in cases where the combined Decay-Precession (DP) model is not statistically preferred, it can still yield more accurate parameter recovery than the simpler models. This occurs because the DP model captures both physical effects present in the data, whereas the simpler models attempt to approximate the signal using only a subset of the underlying physics. However, despite this improved accuracy, the DP model is often penalized in Bayesian evidence due to its increased complexity and larger parameter space.

Furthermore, in regimes where both the orbital decay rate and the apsidal precession rate are large, the parameter space becomes increasingly difficult to constrain. In these cases, the likelihood surface develops strong parameter correlations and broad regions of comparable probability, which can hinder convergence of the nested sampling algorithm. This reflects the fact that rapidly varying secular and periodic signals can become difficult to disentangle, even with relatively large datasets.

This highlights an important distinction between model selection and parameter accuracy: a model that is statistically preferred is not necessarily the one that most accurately reflects the true underlying physical processes, particularly in regimes where the parameter space is poorly constrained.

eccentricity	true_PdE	true_WdE	fit_decay_P	fit_prec_wdE	fit_dp_PdE	fit_dp_wdE	logZ_decay	logZ_prec	logZ_dp
1.00E-04	0.00E+00	0.00E+00	-4.39E-12	-7.56E-09	-4.81E-12	4.82E-09	-6.15E+02	-6.08E+02	-6.13E+02
1.00E-04	0.00E+00	1.00E-08	1.85E-11	5.05E-10	1.94E-11	-2.28E-08	-6.24E+02	-6.18E+02	-6.24E+02
1.00E-04	0.00E+00	1.00E-07	2.38E-11	-1.60E-08	2.22E-11	-9.87E-09	-6.29E+02	-6.25E+02	-6.29E+02
1.00E-04	0.00E+00	1.00E-06	-1.42E-11	3.15E-07	-1.71E-11	-1.43E-07	-6.15E+02	-6.09E+02	-6.15E+02
1.00E-04	0.00E+00	1.00E-05	8.03E-12	-2.46E-05	2.23E-12	-7.21E-06	-5.98E+02	-5.91E+02	-5.98E+02
1.00E-04	0.00E+00	1.00E-04	1.32E-11	-2.19E-04	1.00E-11	-2.29E-04	-6.16E+02	-6.09E+02	-6.15E+02
1.00E-04	-1.00E-12	0.00E+00	-8.19E-12	-1.59E-08	-7.99E-12	1.06E-08	-6.10E+02	-6.04E+02	-6.10E+02
1.00E-04	-1.00E-12	1.00E-08	6.93E-12	1.06E-08	1.36E-11	1.07E-09	-5.82E+02	-5.77E+02	-5.83E+02
1.00E-04	-1.00E-12	1.00E-07	-3.15E-13	2.08E-07	3.62E-12	-2.52E-07	-6.06E+02	-5.99E+02	-6.05E+02
1.00E-04	-1.00E-12	1.00E-06	1.32E-12	1.51E-07	-8.23E-13	-9.42E-07	-6.26E+02	-6.19E+02	-6.26E+02
1.00E-04	-1.00E-12	1.00E-05	1.99E-11	-5.65E-05	1.44E-11	-2.08E-05	-5.88E+02	-5.82E+02	-5.88E+02
1.00E-04	-1.00E-12	1.00E-04	-2.12E-13	-2.29E-04	-3.12E-13	-2.20E-04	-5.65E+02	-5.56E+02	-5.62E+02
1.00E-04	-1.00E-11	0.00E+00	-4.73E-11	-9.27E-09	-5.16E-11	-4.05E-09	-5.79E+02	-5.81E+02	-5.79E+02
1.00E-04	-1.00E-11	1.00E-08	-3.08E-11	-8.31E-09	-3.60E-11	-1.63E-08	-5.92E+02	-5.91E+02	-5.92E+02
1.00E-04	-1.00E-11	1.00E-07	-2.90E-11	2.74E-07	-3.25E-11	-2.18E-07	-5.94E+02	-5.91E+02	-5.94E+02
1.00E-04	-1.00E-11	1.00E-06	-1.18E-11	-2.07E-06	-1.07E-11	1.62E-08	-5.81E+02	-5.75E+02	-5.82E+02
1.00E-04	-1.00E-11	1.00E-05	-3.61E-12	-6.89E-05	9.50E-13	-5.52E-05	-6.46E+02	-6.39E+02	-6.45E+02
1.00E-04	-1.00E-11	1.00E-04	-2.25E-11	-2.84E-04	-1.44E-11	-3.11E-04	-6.35E+02	-6.26E+02	-6.30E+02
1.00E-04	-1.00E-10	0.00E+00	-2.07E-10	1.01E-08	-2.10E-10	5.01E-09	-6.07E+02	-7.43E+02	-6.08E+02
1.00E-04	-1.00E-10	1.00E-08	-2.16E-10	-1.03E-08	-2.23E-10	6.16E-10	-6.12E+02	-7.53E+02	-6.11E+02
1.00E-04	-1.00E-10	1.00E-07	-2.07E-10	-6.71E-08	-2.04E-10	1.47E-08	-6.31E+02	-7.59E+02	-6.31E+02
1.00E-04	-1.00E-10	1.00E-06	-1.89E-10	-4.05E-07	-1.90E-10	-1.14E-06	-7.02E+02	-8.17E+02	-7.02E+02
1.00E-04	-1.00E-10	1.00E-05	-2.15E-10	-6.90E-05	-2.14E-10	-5.38E-05	-5.80E+02	-7.23E+02	-5.80E+02
1.00E-04	-1.00E-10	1.00E-04	-2.08E-10	-1.38E-04	-2.01E-10	-1.60E-04	-6.51E+02	-7.78E+02	-6.51E+02
1.00E-04	-1.00E-09	0.00E+00	-2.00E-09	-1.09E-09	-1.99E-09	3.27E-08	-6.53E+02	-1.36E+04	-6.53E+02
1.00E-04	-1.00E-09	1.00E-08	-2.00E-09	1.42E-08	-2.00E-09	6.25E-09	-5.87E+02	-1.31E+04	-5.86E+02
1.00E-04	-1.00E-09	1.00E-07	-2.01E-09	-1.14E-07	-2.01E-09	8.50E-08	-6.40E+02	-1.31E+04	-6.40E+02
1.00E-04	-1.00E-09	1.00E-06	-2.03E-09	6.08E-07	-2.03E-09	-5.63E-07	-6.22E+02	-1.33E+04	-6.21E+02
1.00E-04	-1.00E-09	1.00E-05	-2.01E-09	-3.31E-05	-2.02E-09	-6.57E-05	-5.72E+02	-1.29E+04	-5.71E+02
1.00E-04	-1.00E-09	1.00E-04	-1.98E-09	-1.12E-04	-1.99E-09	-2.70E-04	-6.28E+02	-1.31E+04	-6.26E+02
1.00E-04	-1.00E-08	0.00E+00	-2.10E-08				-1.36E+03		
1.00E-04	-1.00E-08	1.00E-08	-2.09E-08				-1.27E+03		

Figure 3.1: Representative nested sampling results for $e = 10^{-4}$ with 1000 data points. The table shows the true input parameters, recovered parameters for each model, and the corresponding Bayesian evidence values ($\ln \mathcal{Z}$). Cells are color-coded according to the preferred model and the strength of evidence: green indicates strong preference ($\Delta \ln \mathcal{Z} > 5$) for the precession model, yellow indicates weak preference ($\Delta \ln \mathcal{Z} < 5$) for precession, red indicates weak preference for the decay model, and blue indicates weak preference for the combined DP model. Grey entries correspond to cases where the fitting procedure failed to converge or produced unreliable results.

Chapter 4

Discussion

4.1 Implications for Model Selection

Transit timing variation analyses have traditionally treated orbital decay and apsidal precession as separate, single-effect processes. When only one physical mechanism is present, this approach is well-justified, and our results confirm that the corresponding single-effect model can accurately recover the true system parameters.

However, when both orbital decay and apsidal precession occur simultaneously, this assumption breaks down. In such cases, fitting the data with a single-effect model can lead to biased parameter estimates, as the model attempts to compensate for the missing physical contribution. This can result in systematic overestimation or misinterpretation of the dominant effect, particularly when the secondary signal is non-negligible but not easily distinguishable.

Our results demonstrate that under these conditions, the combined Decay-Precession (DP) model provides a more complete and physically consistent description of the system. By incorporating both secular and periodic components, the DP model reduces these biases and better captures the underlying dynamics, even in regimes where statistical evidence alone may not strongly favor its increased complexity.

This reveals a fundamental tension between statistical preference and physical accuracy. While single-effect models are often strictly favored by formal Bayesian evidence due to their lower parameter complexity (a mathematical consequence of Occam's razor), they can objectively fail to recover the true underlying physical state of the system. These findings powerfully emphasize that model selection based solely on statistical preference can actively mislead interpretation, highlighting the urgent need for careful consideration of combined, multi-process assumptions when analyzing high-precision transit timing variations.

4.2 Observational Constraints and Identifiability

A key outcome of this study is that the ability to determine whether a system is undergoing orbital decay, apsidal precession, or both depends strongly on the quantity and relative quality of observational data. For limited datasets, such as the 1000 data point cases, the statistical evidence is often insufficient to strongly distinguish between models, even when the underlying physical processes are known.

As the dataset size increases, the statistical power improves, allowing weaker signals to be resolved and enabling more reliable model selection. This reflects the critical role of long observational baselines in definitively detecting secular trends such as orbital decay over periodic precession modulations.

In addition to baseline length, robust identifiability between competing models is strongly influenced by the availability of secondary eclipse timing data. While primary transit observations are more readily obtained, eclipse measurements provide critical constraints on the orbital geometry and are unequivocally essential for breaking degeneracies between secular and periodic timing variations. However, such observational constraints remain practically challenging

to acquire, particularly for faint or distant exoplanets.

Nevertheless, the persistence of weak evidence in certain highly-correlated parameter regimes suggests that there may be intrinsic limits in disentangling multiple effects when their observational signatures heavily overlap. Taken together, these results highlight that the interpretation of transit timing variations is fundamentally constrained not only by model framework, but heavily by the inherent limits and challenges of observational data collection.

4.3 Limitations of the Present Study

Beyond purely observational challenges, there are also fundamental physical and methodological limitations within the theoretical framework utilized in this study that must be acknowledged.

Chiefly, in this work, orbital decay and apsidal precession are treated phenomenologically as explicitly independent (decoupled) parameters. In realistic dynamical systems, however, these effects may be powerfully coupled through underlying mechanisms such as tidal interactions, which continuously and simultaneously alter both the planet's orbital evolution (decay) and the geometrical orientation (precession rates). Because the theoretical DP model adopted here lacks these physically coupled differential equations, it serves as a leading-order approximation and may not fully capture the complete, long-term dynamical behavior of real exoplanetary architectures.

4.4 Future Work and Physical Implications

The results presented in this work emphasize the need to consider the simultaneous presence of multiple physical processes when interpreting transit timing variations. In particular, the coexistence of orbital decay and apsidal precession has important implications for understanding planetary structure, tidal dissipation, and long-term orbital evolution in close-in exoplanetary systems.

In this study, orbital decay and apsidal precession are treated as independent (decoupled) effects. However, in realistic systems these processes may be intrinsically linked through their shared dependence on the orbital configuration and internal structure of the star–planet system. For example, apsidal precession continuously rotates the argument of periastron, altering the orbital geometry at transit and eclipse. Because tidal dissipation—and thus orbital decay—depends sensitively on orbital separation and phase-dependent tidal forcing, this geometric evolution may modulate the effective strength of tidal interactions over time. In certain regimes, this could lead to a partial suppression or redistribution of the observable decay signal, complicating its interpretation in transit timing data.

Furthermore, the simultaneous detection of decay and precession provides a powerful probe of planetary interior properties. Orbital decay is directly related to tidal dissipation efficiency, often parameterized by the tidal quality factor, which encodes information about the internal structure and energy dissipation mechanisms within the planet and host star. Apsidal precession, on the other hand, is sensitive to the mass distribution and gravitational moments of the system, including contributions from stellar oblateness, tidal deformation, and relativistic effects. Jointly modeling these processes could therefore provide tighter constraints on planetary structure and the physical conditions governing tidal evolution.

Beyond transit timing measurements, complementary observational techniques may further help disentangle these effects. In particular, radial velocity (RV) observations probe the line-of-sight motion of the host star induced by the orbiting planet. While orbital decay primarily

manifests as a secular drift in the orbital period, apsidal precession alters the shape and phase of the RV curve through changes in the argument of periastron. High-precision RV monitoring over long baselines may therefore provide an independent means of detecting or constraining precession, helping to break degeneracies present in transit timing analyses.

Future work should focus on developing more physically complete models that explicitly incorporate coupling between decay and precession, as well as exploring analytic treatments of their combined evolution. On the observational side, improving timing precision and extending temporal baselines will be essential. In particular, secondary eclipse measurements offer a critical constraint due to their sensitivity to orbital geometry, although such observations remain challenging to obtain.

By advancing both theoretical modeling and observational capabilities, it will be possible to gain deeper insight into the complex and interconnected dynamical processes governing close-in exoplanetary systems.

4.5 Conclusion

In this work, we have developed and investigated a unified framework for modeling transit timing variations arising from orbital decay, apsidal precession, and their combined effects. By introducing the Decay-Precession (DP) model and systematically exploring a wide parameter space, we demonstrate that multiple physical processes can simultaneously shape the observed timing signals. This work extends the conventional approach in transit timing analyses, which typically considers these mechanisms independently, by explicitly examining their coexistence and observational consequences.

Our results show that when only a single physical effect is present, the corresponding model accurately recovers the true system parameters, validating the reliability of traditional single-effect approaches in appropriate regimes. However, when orbital decay and apsidal precession occur simultaneously, single-effect models can produce biased or misleading parameter estimates, as they attempt to compensate for missing physical contributions. In contrast, the DP model provides a more complete representation of the system, capturing both the secular and periodic components of the timing signal.

We find that the ability to distinguish between these mechanisms is fundamentally limited by observational coverage and signal strength. In data-limited regimes, single-effect models may still be statistically preferred due to their simplicity, even when they do not fully represent the underlying physics. As the observational baseline increases, the combined DP model becomes increasingly favored, enabling more accurate recovery of the true parameters and revealing the presence of multiple contributing processes.

A key outcome of this work is exposing the fundamental tension between statistical preference and physical accuracy. Our results explicitly show that a model heavily favored by Bayesian evidence (due to parameter simplicity) is not necessarily the one that accurately reflects the true dynamical state of the system. This highlights a critical and often overlooked challenge in exoplanet characterization: the urgent need to balance mathematical model simplicity with genuine physical interpretability, while remaining hyper-aware of the limitations imposed by the current observational baselines.

More broadly, this work emphasizes that transit timing variations are not merely diagnostic tools for isolated processes, but rather probes of complex and potentially coupled dynamical phenomena. The coexistence of orbital decay and apsidal precession has direct implications for our understanding of tidal dissipation, planetary interior structure, and long-term orbital

evolution in close-in exoplanet systems.

Future efforts should focus on developing more physically complete models that account for potential coupling between different mechanisms, as well as improving observational constraints. In particular, secondary eclipse timing measurements offer a powerful means of breaking degeneracies between competing models, although such observations remain challenging. Additionally, further analytic and numerical studies of coupled orbital evolution will be essential for refining theoretical predictions.

Ultimately, recognizing and modeling the simultaneous action of multiple dynamical processes will be crucial for extracting reliable physical insights from the next generation of high-precision exoplanet observations. This work therefore suggests that future interpretations of transit timing data should move beyond single-mechanism assumptions and toward a unified, multi-process framework.

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Appendix A

Stroboscopic DP Simulation Algorithm

The following pseudocode outlines the core stroboscopic logic implemented in `delta_te.py`. It demonstrates how the simulation accurately avoids continuous integration by directly stepping to an observation epoch, modifying the orbital elements with secular decay, calculating the coupled apsidal precession perturbations induced by the stellar oblateness (J_2), and utilizing a secant root-finding proxy to calculate the exact mid-transit passing.

```
def simulate_transit_time(E_target, P_0, PdE, J2, a_initial, M_star, R_star, e, I):
    """
    Computes the exact O-C timing variation for a given discrete epoch E_target.
    """
    # 1. Apply secular orbital decay
    P_actual = P_0 + (PdE * E_target)

    # Calculate shrinking semi-major axis (Keplers 3rd Law proxy)
    a_actual = ((P_actual / (2 * pi))**2 * G * M_star)**(1/3)
    n_actual = sqrt(G * M_star / a_actual**3)

    # 2. Recompute geometry-dependent J2 Perturbations (The Coupling)
    wdot = 0.75 * n_actual * J2 * (R_star/a_actual)**2 * (5*cos(I)**2 - 1) / (1-e**2)**2
    Odot = -1.5 * n_actual * J2 * (R_star/a_actual)**2 * cos(I) / (1-e**2)**2
    Mdot = n_actual + 0.75 * n_actual * J2 * (R_star/a_actual)**2 * (3*cos(I)**2 - 1) / (1-e**2)**1.5

    # 3. Advance orbital angles
    w_E = w_0 + wdot * E_target
    O_E = O_0 + Odot * E_target
    M_E = M_0 + Mdot * E_target

    # 4. Minimize projected sky-plane distance using secant method
    # Evaluate at two test points close to zero to interpolate exact transit root
    dt_1 = -500  # seconds offset
    dt_2 = +500  # seconds offset

    # Evaluate derivative function (2Y*dY + 2Z*dZ)
    dsdf_1 = evaluate_projected_distance_derivative(dt_1, w_E, O_E, M_E)
    dsdf_2 = evaluate_projected_distance_derivative(dt_2, w_E, O_E, M_E)

    # Secant interpolation to find root where derivative == 0
    exact_time_offset = dt_1 - dsdf_1 * (dt_2 - dt_1) / (dsdf_2 - dsdf_1)

    t_actual = (P_0 * E_target) + exact_time_offset
    O_C = t_actual - (P_0 * E_target)
```

```
36
37      return O_C
```

Appendix B

Parameter Sweep and Automated Workflow Algorithm

To systematically map the parameter limits of the competing physical mechanisms and track the Bayesian evidential response across different timing signal strengths, a high-level automated workflow script (`automate_workflow.py`) was utilized. This appendix provides the pseudocode detailing the outer parameter loop, synthetic data generation, constraint modification, and dynamic execution of the nested sampling framework.

```
def execute_large_scale_simulation(e_values, PdE_values, WdE_target_values):
    """
    Executes a multi-dimensional parameter sweep bridging delta_te.py
    generators with the orbdot nested sampling analysis.
    """
    # Grid search across all combinations of physical parameters
    for e in e_values:
        for PdE in PdE_values:
            for WdE_target in WdE_target_values:

                # 1. Derive matching physical J2 parameter for target WdE
                J2_derived = calculate_J2_from_WdE(WdE_target, e, a_initial,
                    M_star, R_star)

                # 2. Modify delta_te.py runtime configuration
                modify_simulation_config(e, J2_derived, PdE)

                # 3. Execute stroboscopic simulation
                raw_data = run_stroboscopic_generator()

                # 4. Inject Gaussian measurement uncertainties
                simulated_data = inject_gaussian_noise(raw_data,
                    base_uncertainty=0.0004)

                # 5. Dynamically widen prior boundaries based on injected
                    parameters
                generate_orbdot_settings(e, PdE, WdE_target)

                # 6. Execute rigorous Bayesian model comparison via nested
                    sampling
                try:
                    fit_decay = run_orbdot_nested_sampling(model="decay")
                    fit_prec  = run_orbdot_nested_sampling(model="precession")
                    fit_dp    = run_orbdot_nested_sampling(model="
                        decay_precession")

                    # 7. Extract the Bayesian Evidence (Z) for model comparison
                    record_to_csv([e, PdE, WdE_target,
                                   fit_decay.logZ, fit_prec.logZ, fit_dp.logZ])

```

```
36                 except TimeoutException:
37                     handle_failure_and_retry(scale_factor=10)
```